

Answer Assistant, answer multiple-choice questions and point to the answer

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Introduction

In this tutorial, we will demonstrate how to use the Dify interface to chat with Muto, allowing it to observe the question in front of it and point to the answer.

Note that Muto can only raise its left and right front legs, so we recommend having Muto face true/false questions or multiple-choice questions with only two options.

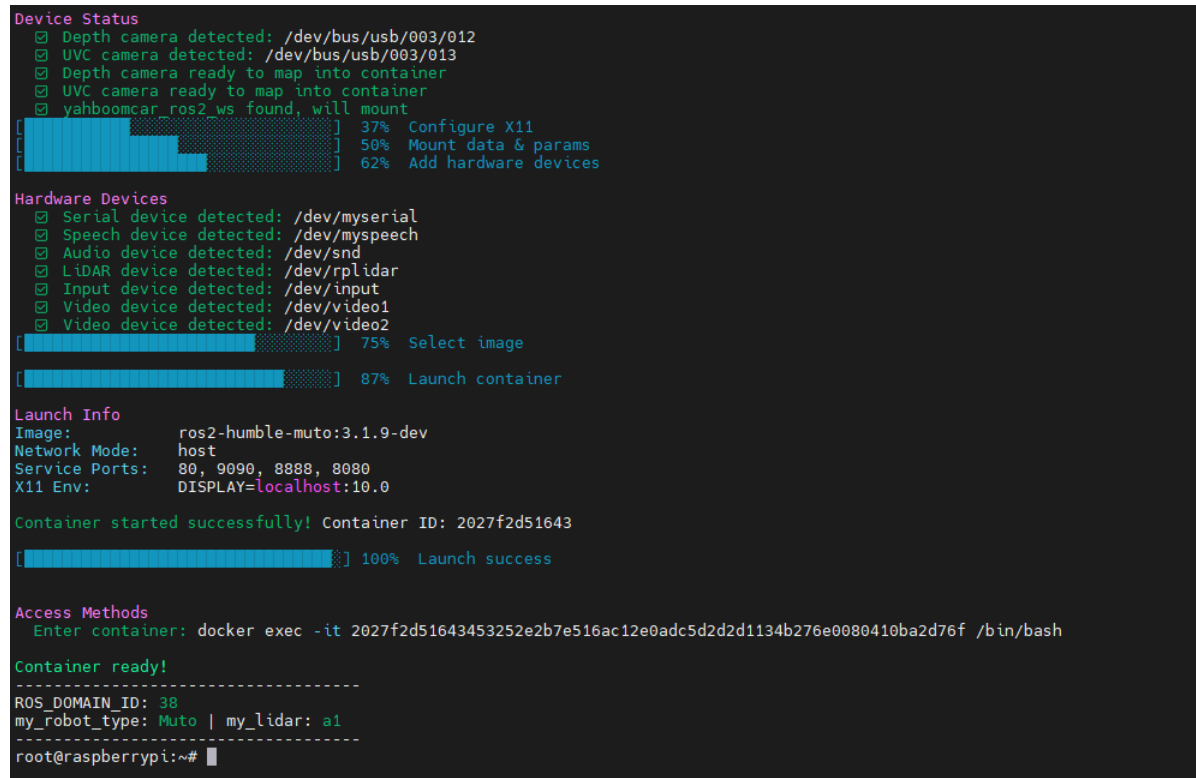
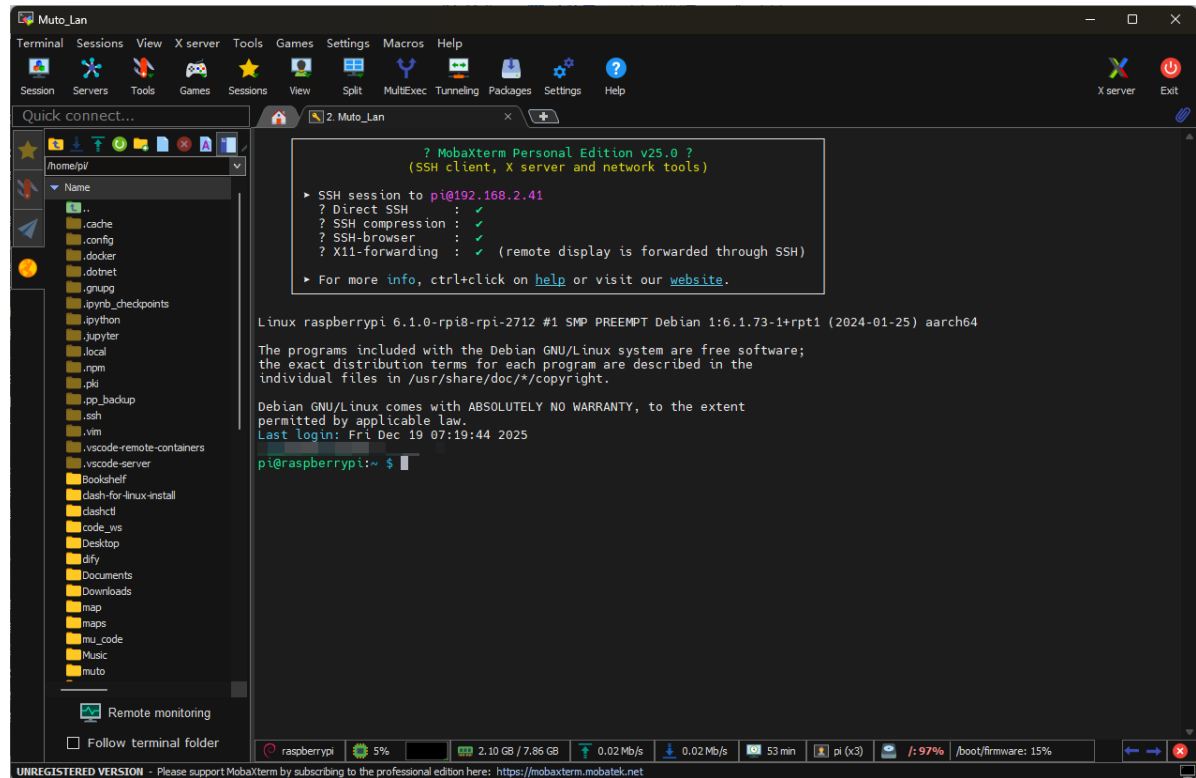
Operation Steps

Preparatory Steps

This lesson requires the Dify terminal to be running, ensuring Dify is functioning correctly and all configuration items are set up as described in previous sections.

The screenshot displays the Dify Studio interface for a workflow named 'MutoLLM2.0_EN'. The top navigation bar includes 'Explore', 'Studio', 'Knowledge', and 'Tools'. The left sidebar shows 'Orchestrate', 'API Access', 'Logs & Annotations', and 'Monitoring'. The main workspace shows a workflow diagram with nodes: 'IS HAVE A LOOK' (triggered by 'USER L...'), 'HAVE A LOOK' (using 'qwen-vl-max'), 'ANSWER' (using 'qwen-vl-max'), 'DECISION_LAYER_PROMPT', 'DECISION' (using 'qwen-vl-max'), 'EXECUTION_LAYER_PROMPT', 'EXECUTION_LAYER_PROMPT', 'PRE-PROCESSING', and 'ANSWER 3'. The right sidebar shows the 'Publish' button and a 'Publish Update' button. The bottom right panel shows the 'SYSTEM' role definition and core tasks for the 'Hexapod Robot Decision Layer Control System'.

This lesson requires the MutoRS to be powered on. Follow the instructions in the previous sections... Starting a large model terminal in Docker



```

[muto_controller_node-1] 2025-12-19 09:48:12,313 - voice_module.voice_interface - INFO - Speech to text initialized successfully with alibaba provider
[muto_controller_node-1] 2025-12-19 09:48:12,313 - voice_module.core.text_to_speech - INFO - TTS Provider alibaba
[muto_controller_node-1] 2025-12-19 09:48:12,314 - voice_module.core.audio_queue_manager - INFO - Audio queue processor started
[muto_controller_node-1] 2025-12-19 09:48:12,314 - voice_module.core.audio_queue_manager - INFO - Audio queue processor started
[muto_controller_node-1] 2025-12-19 09:48:12,314 - voice_module.voice_interface - INFO - Audio queue manager initialized successfully
[muto_controller_node-1] 2025-12-19 09:48:12,314 - voice_module.voice_interface - INFO - All voice modules initialized successfully
[muto_controller_node-1] [INFO] [1766137692.315363643] [muto_controller_node]: Command executor initialized successfully
[muto_controller_node-1] [INFO] [1766137692.315920470] [muto_controller_node]: Attempting to prestart camera system...
[muto_controller_node-1] X Node not running: /camera/camera
[muto_controller_node-1] Current running nodes: ['/muto_controller_node', '/navigation_manager']
[muto_controller_node-1] [INFO] [1766137697.078459054] [muto_controller_node]: Camera service started successfully: Camera started successfully
[muto_controller_node-1] [INFO] [1766137697.085229984] [persistent_image_capture_3188dc8d]: Persistent image capture node initialized for topic: /camera/color/image_raw with Best Effort QoS
[muto_controller_node-1] [INFO] [1766137697.086642747] [muto_controller_node]: Persistent camera node initialized: Camera node initialized successfully
[muto_controller_node-1] [INFO] [1766137697.087635385] [muto_controller_node]: FastAPI application created successfully
[muto_controller_node-1] [INFO] [1766137697.149655757] [muto_controller_node]: API routes registered successfully
[muto_controller_node-1] 2025-12-19 09:48:17,149 - voice_module.core.voice_wake - INFO - Starting voice wake detector on port: /dev/myspeech
[muto_controller_node-1] 2025-12-19 09:48:17,154 - utils.serial_comm - INFO - Serial data receiving started
[muto_controller_node-1] 2025-12-19 09:48:17,154 - voice_module.core.voice_wake - INFO - Voice wake detector started successfully
[muto_controller_node-1] 2025-12-19 09:48:17,154 - voice_module.voice_interface - INFO - Wake detection started
[muto_controller_node-1] [INFO] [1766137697.155236161] [muto_controller_node]: Voice assistant auto-start: True
[muto_controller_node-1] [INFO] [1766137697.156658443] [muto_controller_node]: ROS2 publisher created successfully
[muto_controller_node-1] [INFO] [1766137697.157379009] [muto_controller_node]: Status timer and health check created successfully
[muto_controller_node-1] [INFO] [1766137697.157921559] [muto_controller_node]: Muto Controller Node initialized successfully
[muto_controller_node-1] [INFO] [1766137697.158425850] [muto_controller_node]: Starting FastAPI server on 0.0.0.0:8080
[muto_controller_node-1] [INFO] [1766137697.161239210] [muto_controller_node]: FastAPI server started successfully
[muto_controller_node-1] [INFO]: Started server process [221]
[muto_controller_node-1] [INFO]: Waiting for application startup.
[muto_controller_node-1] [INFO]: Application startup complete.
[muto_controller_node-1] [INFO]: Uvicorn running on http://0.0.0.0:8080 (Press CTRL+C to quit)

```

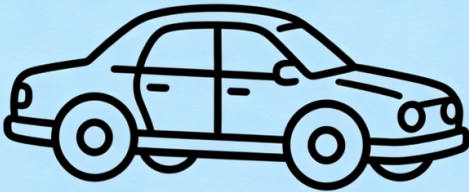
Getting Started

The command words we will use in this lesson are as follows:

Look at the multiple-choice question in front of you. Please analyze it for me and then point to the correct answer.

The question we will use is as follows:

A car travels at a constant speed of 60km/h for the 2 hours.



what is the total distance traveled ?

Distance=Speed×Time

A. 30 km B. 120 km

Muto's execution flow is as follows:

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Orchestrate

API Access

Logs & Annotations

Monitoring

Workflow

IF/ELSE 4

IF

Get JSON

ELSE

END2

ANSWER

Parse Content

Play Voice

PREVIEW

image_file (optional)

request_id (optional)

TEXT_VERSION

ip_address

192.168.12.49

Look at the multiple-choice question in front of you. Please analyze it for me and then point to the correct answer.

Auto-Saved 04:09:34 - Published a day ago

Orchestrate

API Access

Logs & Annotations

Monitoring

Workflow

IF/ELSE 4

Get JSON

ELSE

END2

ANSWER

Parse Content

Play Voice

PREVIEW

Look at the multiple-choice question in front of you. Please analyze it for me and then point to the correct answer.

Workflow Process

Step	Duration	Status
USER INPUT	9.379 ms	Success
IS HAVE A LOOK	71.907 ms	Success
DECISION_LAYER_PROMPTS	7.703K tokens - 4.307 s	Success
DECISION	44.599 ms	Success
EXECUTION_LAYER_PROMPTS	16.606K tokens - 18.399 s	Success
JUDGEMENT_BROADCAST	741 tokens - 1.050 s	Success
PARSE JSON	81.399 ms	Success
GET JSON	62.347 ms	Success
IF/ELSE 4	40.662 ms	Success
END2	31.688 ms	Success

The result of this execution is as follows

- (Start intelligent visual analysis and identify the multiple-choice question content in front, analyze the question and determine the correct answer): (B)
- (Original Action Requirement): Walk forward three steps and broadcast the detailed analysis result, then if the correct answer is option A, execute point left front action; if it is option B, execute point right front action (Compressed Execution Result, Data): Success: true, Commands queued for async execution, Execution time: 5.34e-05 seconds

Talk to Bot

Observing the terminal, you can see that Muto has carefully analyzed the question and provided a detailed explanation.

```

Health check endpoint: http://0.0.0.0:8080/health
API documentation: http://0.0.0.0:8080/docs
[INFO] [muto_controller_node-1]: process started with pid [776]
[muto_controller_node-1] Initializing Muto Controller Node...
[muto_controller_node-1] Navigation Manager ROS2 components initialized
[muto_controller_node-1] VoiceInterface initialized with region: china, language: en, STT provider:
[muto_controller_node-1] Voice wake detector initialized
[muto_controller_node-1] Voice recorder initialized
[muto_controller_node-1] aplay verified: aplay
[muto_controller_node-1] Initializing STT with credentials: AppID=f12672f1, APIKey=8c7b***, APISec
[muto_controller_node-1] STT Provider xunfei
[muto_controller_node-1] Speech to text initialized successfully with xunfei provider
[muto_controller_node-1] TTS Provider xunfei
[muto_controller_node-1] Audio queue processor started
[muto_controller_node-1] Audio queue manager initialized successfully
[muto_controller_node-1] All voice modules initialized successfully
[muto_controller_node-1] Command executor initialized successfully
[muto_controller_node-1] Attempting to prestart camera system...
[muto_controller_node-1] Camera service started successfully: Camera started successfully
[muto_controller_node-1] Persistent image capture node initialized for topic: /camera/color/image
[muto_controller_node-1] Persistent camera node initialized: Camera node initialized successfully
[muto_controller_node-1] FastAPI application created successfully
[muto_controller_node-1] API routes registered successfully
[muto_controller_node-1] Starting voice wake detector on port: /dev/myspeech
[muto_controller_node-1] Serial data receiving started
[muto_controller_node-1] Voice wake detector started successfully
[muto_controller_node-1] Wake detection started
[muto_controller_node-1] ROS2 publisher created successfully
[muto_controller_node-1] Status timer and health check created successfully
[muto_controller_node-1] Starting FastAPI server on 0.0.0.0:8080
[muto_controller_node-1] FastAPI server started successfully
[muto_controller_node-1] INFO: Started server process [776]
[muto_controller_node-1] INFO: Waiting for application startup.
[muto_controller_node-1] INFO: Application startup complete.
[muto_controller_node-1] INFO: uvicorn running on http://0.0.0.0:8080 (Press CTRL+C to quit)
[muto_controller_node-1] Mode: Sync (Blocking)
[muto_controller_node-1] Plan Data: [{"robot_speak(text='I am analyzing the question in front of me.')}", "have_a_look(user_query='Identify the multiple-choice question in view, analyze its content, and determine the correct answer. Return the answer as either 'A' or 'B' based on the options.')]
[muto_controller_node-1] [Muto reply] I am analyzing the question in front of me.
[muto_controller_node-1] [event]: "execute_commands", "success": true, "executed_count": 3, "failed_command_id": null, "duration_ms": 5884}
[muto_controller_node-1] Mode: Async (Background)
[muto_controller_node-1] Plan Data: [{"robot_speak(text='The correct answer is option B.')}", "forward(steps=3)}", "point_right_front()"}]
[muto_controller_node-1] [Muto reply] The correct answer is option B.
[muto_controller_node-1] [DEBUG] 执行移动动作: forward, 距离=None, 速度=None, 步数=3, 时间=None
[muto_controller_node-1] [DEBUG] 配置文件中指定10的速度: 0.02919103632731524 m/s
[muto_controller_node-1] [DEBUG] 根据步数计算执行时间: 3 steps * 0.5 s/step = 1.5 s
[muto_controller_node-1] [DEBUG] 开始执行移动, 预计执行时间: 1.5 s
[muto_controller_node-1] [DEBUG] 移动执行完成, 实际执行时间: 1.994 s
[muto_controller_node-1] [DEBUG] move(0, 0, 0) test

```

feh [1 of 1] - /tmp/camera_capture_1769072983.jpg@raspberrypi

A car travels at a constant speed of 60km/h for the 2 hours. what is the total distance traveled ?

Distance=Speed×Time

A. 30 km B. 120 km

